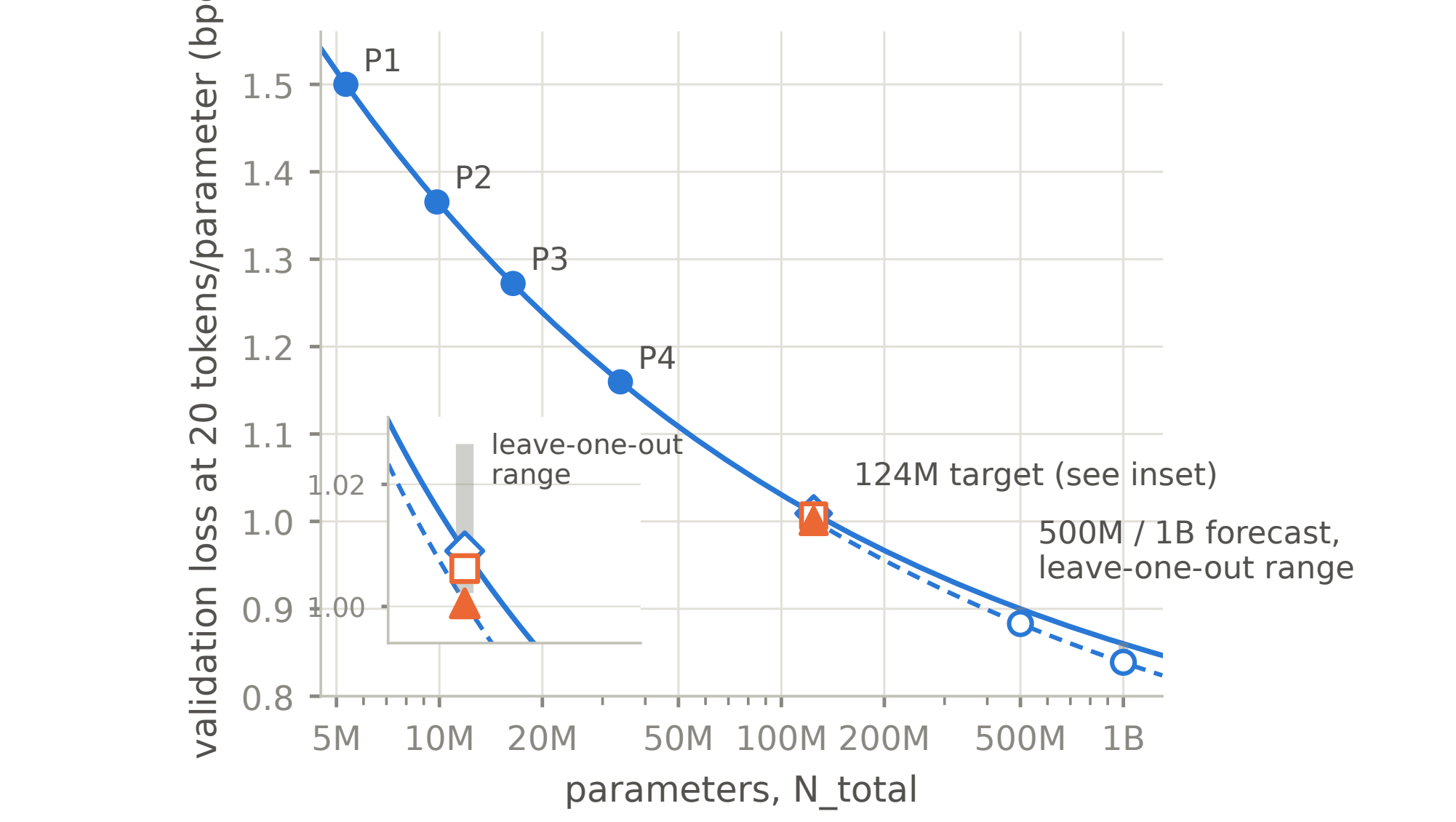
validation loss at 20 tokens/parameter (bpb)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
P1
P2
P3
P4
leave-one-out range
1.02
1.00
124M target (see inset)
500M / 1B forecast, leave-one-out range
5M
10M
20M
50M
100M
200M
500M
1B
parameters, N_total
registered four-rung ray
five-rung refit (§8 forecast)
ladder rungs, 20× (masked)
registered prediction 1.0091
measured, branch of the 80×-rate trunk 1.0062
measured, own 20× rate 1.0005